

Table A.9: Performance of C–G (Action Program) under two randomization controls. **Categories Shuffle** randomly permutes the assignment of support examples to positive vs. negative sets within each problem, while leaving the query and its gold label unchanged. **Test Sequence Shuffle** randomly shuffles the order of drawing actions in the query program at inference time, leaving the support examples, query image, and gold label unchanged. Values show accuracy (%) on Basic (BD) and Free-form (FF) problems for each model.

Model	Shuffle Type	BD	FF
DeepSeek-R1:14B	Categories	63.0	63.0
	Test Sequence	61.6	52.2
DeepSeek-R1:32B	Categories	58.8	73.0
	Test Sequence	68.2	68.8
DeepSeek-R1:8B	Categories	54.3	54.0
	Test Sequence	54.6	51.4
Gemma3:12B	Categories	57.6	63.2
	Test Sequence	64.7	54.2
Gemma3:27B	Categories	58.6	76.6
	Test Sequence	65.0	55.4
Magistral	Categories	64.2	70.9
	Test Sequence	66.4	61.4
Phi-4	Categories	61.4	81.2
	Test Sequence	60.0	53.4
Phi-4-Reasoning	Categories	57.8	86.8
	Test Sequence	65.2	52.6
Qwen2.5:32B	Categories	63.8	54.6
	Test Sequence	71.6	67.8

Table A.10: Mean accuracy (%) $\pm$ standard deviation across the nine models in Table A.9 for each randomization control. BD = Basic shapes, FF = Free-form shapes.

Shuffle Type	BD	FF
Categories Shuffle	59.9 $\pm$ 3.3	69.3 $\pm$ 11.4
Test Sequence Shuffle	64.1 $\pm$ 4.9	57.5 $\pm$ 6.8

- Formal, unambiguous rules rather than statistical regularities

Results on this benchmark should not be interpreted as evidence of general visual reasoning capability. Evaluation on natural-image Bongard datasets (e.g., Bongard-OpenWorld with VLM–LLM interfaces) and other diverse benchmarks is essential for establishing broader generalization.

F.2 Ground-Truth Privilege and Practical Applicability

Our approach assumes access to perfect symbolic specifications—the ground-truth action programs used to generate each image. This represents a significant privilege that is unavailable in most real-world scenarios:

- Most domains lack formal generative specifications
- Obtaining symbolic descriptions requires either domain-specific engineering, learned perceptual systems, or human annotation

We caution against over-interpreting the success of the C–G approach as evidence that human-level visual reasoning is imminent. Rather, it illustrates what becomes possible when perception is perfectly decoupled from reasoning.

F.3 Model Coverage Gaps

Our evaluation covered recent reasoning-capable models but excluded:

- Vision-language models in their symbolic reasoning mode (architectural constraints limited multi-image symbolic reasoning)
- Older and specialized models (e.g., earlier GPT/Claude variants, mathematical reasoners) were not evaluated systematically.
- Models from certain organizations (OpenAI, Anthropic APIs were not used)
- Smaller models (<7B parameters) were not systematically evaluated

Results on a broader model portfolio would strengthen generality claims and establish baseline expectations for future work.

F.4 Environmental and Computational Impact

This research required:

- €1000 in API costs to run frontier models (financial barrier to reproducibility)
- 2000+ GPU-hours (computational barrier)
- Significant energy consumption (environmental impact)

These barriers limit reproducibility and raise equity concerns.

F.5 Incomplete Analysis

Several aspects of this work merit deeper investigation:

- Human-designed performance ($\approx 60\text{--}66\%$) remains substantially lower than the BD and FF regimes despite perfect symbolic input.
- Robustness to symbolic input corruption (e.g., noise in programs or descriptions) has not been evaluated.

Further work is needed to analyze failure modes in output parsing and to characterize computational efficiency and scaling behavior.